

Unit 1 Assessment*Equations and Inequalities*

Name _____

Date _____

Period _____

Score _____ / 21

DIRECTIONS. Show your work for every question. Write repeating decimals with a bar over the digits that repeat. Fractions do not have to be in simplest form. Where a question gives units, include them in your answer.

1. Complete the table by writing the equivalent value in each empty cell.

3 points

	Fraction	Decimal	Percent
a.	6/11		
b.		0.85	
c.			72%

2. Maria opens a package of Tropical Skittles. The package has five flavors: Banana Berry (yellow), Kiwi Lime (green), Mango Tangelo (orange), Pineapple Passion fruit (blue) and Strawberry Starfruit (pink). The table shows what part of Maria's package is each flavor.

4 points

Flavor	Amount
Banana Berry	20%
Kiwi Lime	$\frac{4}{15}$
Mango Tangelo	0.12
Pineapple Passion fruit	$19.\overline{1}\%$
Strawberry Starfruit	$0.\overline{2}$

- a. What percentage of Maria's package is Mango Tangelo?

1 point

Percentage	
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- b. Rewrite the amount of Maria's package that is Kiwi Lime as a decimal.

1 point

Decimal	
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- c. What fraction of Maria's package is Strawberry Starfruit?

1 point

Fraction	
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- d. Rewrite the amount of Maria's package that is Banana Berry as a decimal.

1 point

Decimal	
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3. Esmeralda is saving for a trip to Siesta Key. She is paid the same amount every week. Each week she uses \$382 of her pay for her expenses and saves everything that is left. After 11 weeks she has saved \$506.

2 points

- a. Write an equation that can be used to determine the amount, x , in dollars that Esmeralda is paid each week.

1 point

Equation	
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- b. Solve the equation.

1 point

Solution	
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Solve each equation. Show your work.

4. $-4(w + 7) = 108$

1 point

Solution	
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5. $n \div (-3) + 3.8 = -7.4$

1 point

Solution	
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6. $3\frac{5}{6} = 8\frac{1}{5} - k$

1 point

Solution	
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7. A server at The Main Street Grill earns \$7.80 per hour plus \$87.40 in tips.

2 points

- a. Write an equation that can be used to determine the number of hours, h , the server worked to earn \$136.15.

1 point

Equation	
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- b. Solve the equation.

1 point

Solution	
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8. Circle every value below that is a solution to $-3x + 4 \leq 19$.

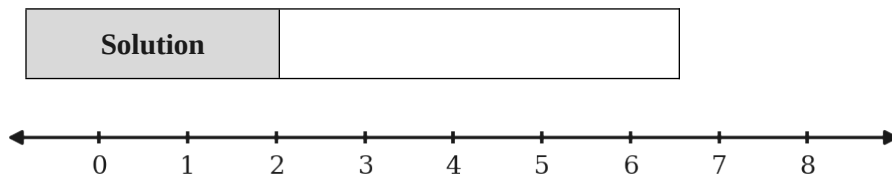
1 point

-8	$-6\frac{1}{3}$	-5	-4.9	0	12.5
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Solve each inequality. Show your work, then graph the solution set on the number line.

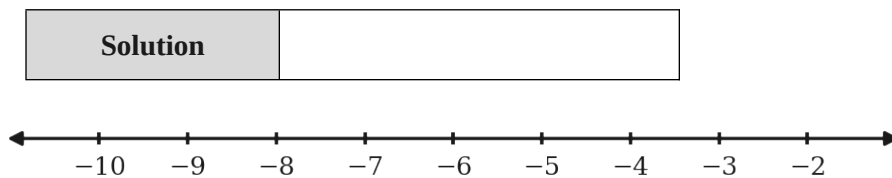
9. $-11.3 > -3x - 0.5$

1 point



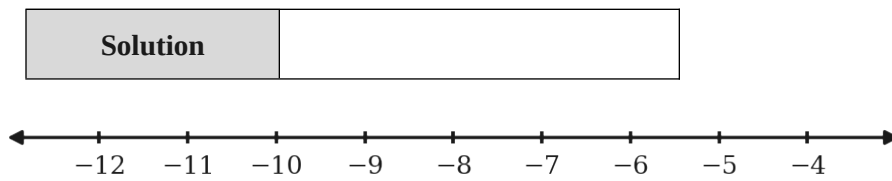
10. $\frac{2}{3}(21 - 15x) > 76$

1 point



11. $-8(5x + 9) \geq 208$

1 point



12. The drama club is renting a bus for a field trip. The bus company charges a flat fee of \$260 plus \$16 for each student who rides. The club has raised \$1,000 and wants to take as many students as possible.

3 points

- a. Write an inequality that can be used to determine the number of students, s , who can ride the bus.

1 point

Inequality	
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- b. Solve the inequality.

1 point

Solution	
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- c. How many students can ride the bus? Justify your answer mathematically.

1 point